

1 Summary

Model (Run-name): model3 (test)

Iteration: 14

TAC: 64106.608845642084

Took: 2181.1185653209686s

Total Time: 4765.628186702728s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

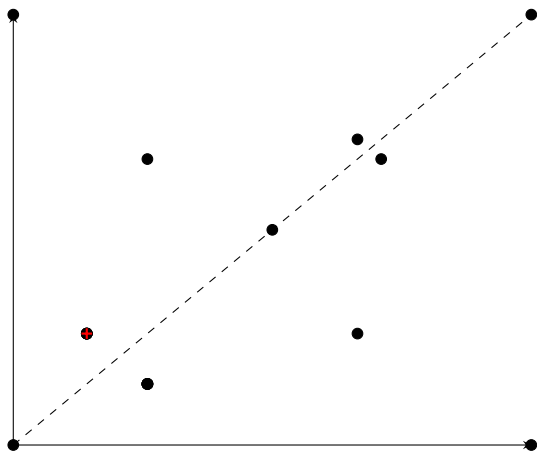
2 Results

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	1.0	$dt_{3,5,1}$	22.3	$q_{1,1,2}$	586.96
$z_{2,2,1}$	1.0	$dt_{3,5,2}$	29.056081081081047	$q_{2,2,1}$	647.2099999999999
$z_{2,4,2}$	1.0	$dt_{4,3,2}$	50.0	$q_{2,4,2}$	530.45
$z_{3,5,1}$	1.0	$dt_{4,3,3}$	111.1	$q_{3,5,1}$	1544.2900000000002
$z_{4,3,2}$	1.0	$dt_{5,1,1}$	38.9	$q_{4,3,2}$	1539.72
$z_{5,1,1}$	1.0	$dt_{5,1,2}$	51.89214986619177	$q_{5,1,1}$	173.03999999998632
$z_{5,4,2}$	1.0	$dt_{5,4,2}$	12.423728813560103	$q_{5,4,2}$	1104.4
$z_{cu,3}$	1.0	$dt_{5,4,3}$	44.52824858757057	$q_{cu,3}$	839.9899999999994
$z_{cu,5}$	1.0	$dt_{cu,3}$	82.35608108108104	$q_{cu,5}$	1081.9699999999993
$A_{1,1,2}^\beta$	6.886621175724489	$dt_{cu,5}$	86.72824858757058	$t_{1,1}^{(H)}$	160.0
$A_{2,2,1}^\beta$	5.378842799033672	$f_{1,1,2}^H$	8.8	$t_{1,2}^{(H)}$	160.0
$A_{2,4,2}^\beta$	4.5903997068045905	$f_{2,2,1}^H$	5.825472547254725	$t_{1,3}^{(H)}$	93.3
$A_{3,5,1}^\beta$	12.923002798326532	$f_{2,4,2}^H$	10.6	$t_{2,1}^{(H)}$	248.9
$A_{4,3,2}^\beta$	6.675778583751798	$f_{3,5,1}^H$	9.5859093730602	$t_{2,2}^{(H)}$	187.8424528301873
$A_{5,1,1}^\beta$	2.128002147530601	$f_{4,3,2}^H$	12.6	$t_{2,3}^{(H)}$	137.8
$A_{5,4,2}^\beta$	10.6608373431196	$f_{5,1,1}^H$	1.298124531132646	$t_{3,1}^{(H)}$	226.7
$A_{cu,3}^\beta$	5.4227983074998	$f_{5,4,2}^H$	8.940792272122206	$t_{3,2}^{(H)}$	122.35608108108104
$A_{cu,5}^\beta$	6.203640738497231	$f_{1,1,2}^C$	5.869600000000002	$t_{3,3}^{(H)}$	122.35608108108104
$A_{1,1,2}$	24.927400386350968	$f_{2,2,1}^C$	6.1	$t_{4,1}^{(H)}$	271.1
$A_{2,2,1}$	16.51257309583564	$f_{2,4,2}^C$	5.613227513227356	$t_{4,2}^{(H)}$	271.1
$A_{2,4,2}$	22.56363950048211	$f_{3,5,1}^C$	13.9	$t_{4,3}^{(H)}$	148.9
$A_{3,5,1}$	71.16597394333012	$f_{4,3,2}^C$	8.4	$t_{5,1}^{(H)}$	198.9
$A_{4,3,2}$	23.668654733871367	$f_{5,1,1}^C$	7.6	$t_{5,2}^{(H)}$	189.1237288135601
$A_{5,1,1}$	4.425807501201696	$f_{5,4,2}^C$	11.686772486772645	$t_{5,3}^{(H)}$	126.72824858757058
$A_{5,4,2}$	51.6615457964989	$RecLMTD_{1,1,2}$	0.03609835812479837	$t_{1,1}^{(C)}$	160.0
$A_{cu,3}$	16.738084386191687	$RecLMTD_{2,2,1}$	0.021686449732638335	$t_{1,2}^{(C)}$	137.23157894736832
$A_{cu,5}$	20.944873037188618	$RecLMTD_{2,4,2}$	0.03615627029014946	$t_{1,3}^{(C)}$	60.0
$dt_{1,1,2}$	22.768421052631616	$RecLMTD_{3,5,1}$	0.039170802020236324	$t_{2,1}^{(C)}$	221.7
$dt_{1,1,3}$	33.3	$RecLMTD_{4,3,2}$	0.01306624355323738	$t_{2,2}^{(C)}$	115.6
$dt_{2,2,1}$	27.2	$RecLMTD_{5,1,1}$	0.0217402703191334	$t_{2,3}^{(C)}$	115.6
$dt_{2,2,2}$	72.24245283018732	$RecLMTD_{5,4,2}$	0.039761240426498014	$t_{3,1}^{(C)}$	221.1
$dt_{2,4,2}$	11.142452830187295	$RecLMTD_{cu,3}$	0.01693754893303839	$t_{3,2}^{(C)}$	221.1
$dt_{2,4,3}$	55.6	$RecLMTD_{cu,5}$	0.016454376814154174	$t_{3,3}^{(C)}$	37.8

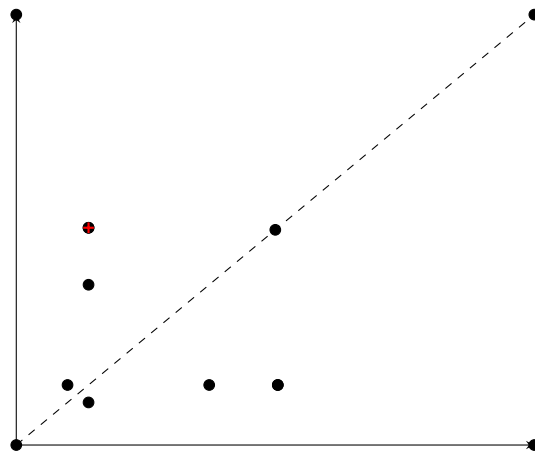
Variable	Value	Variable	Value
$t_{4,1}^{(C)}$	176.7	$b_{5,4,2}^{H,out}$	586.5159730512163
$t_{4,2}^{(C)}$	176.7	$b_{1,1,2}^{C,in}$	352.1760000000001
$t_{4,3}^{(C)}$	82.2	$b_{2,2,1}^{C,in}$	705.16
$t_{5,1}^{(C)}$	204.4	$b_{2,4,2}^{C,in}$	461.40730158728866
$t_{5,2}^{(C)}$	93.3	$b_{3,5,1}^{C,in}$	1296.87
$t_{5,3}^{(C)}$	93.3	$b_{4,3,2}^{C,in}$	317.52
$t_{1,1,2}^H$	93.3	$b_{5,1,1}^{C,in}$	1042.9600000000273
$t_{2,2,1}^H$	138.72781467249376	$b_{5,4,2}^{C,in}$	960.6526984127114
$t_{2,4,2}^H$	137.8	$b_{1,1,2}^{C,out}$	939.1360000000002
$t_{3,5,1}^H$	65.6	$b_{2,2,1}^{C,out}$	1352.37
$t_{4,3,2}^H$	148.9	$b_{2,4,2}^{C,out}$	991.8573015872737
$t_{5,1,1}^H$	65.6	$b_{3,5,1}^{C,out}$	2841.1600000000003
$t_{5,4,2}^H$	65.6	$b_{4,3,2}^{C,out}$	1857.24
$t_{1,1,2}^C$	160.0	$b_{5,1,1}^{C,out}$	1216.0
$t_{2,2,1}^C$	221.7	$b_{5,4,2}^{C,out}$	2065.0526984127264
$t_{2,4,2}^C$	143.00298893996245		
$t_{3,5,1}^C$	204.4		
$t_{4,3,2}^C$	221.1		
$t_{5,1,1}^C$	160.0		
$t_{5,4,2}^C$	176.7		
$b_{1,1,2}^{H,in}$	1408.0		
$b_{2,2,1}^{H,in}$	1449.960117011701		
$b_{2,4,2}^{H,in}$	1991.1299999999924		
$b_{3,5,1}^{H,in}$	2173.1256548727492		
$b_{4,3,2}^{H,in}$	3415.8599999999997		
$b_{5,1,1}^{H,in}$	258.1969692422834		
$b_{5,4,2}^{H,in}$	1690.915973051231		
$b_{1,1,2}^{H,out}$	821.0400000000002		
$b_{2,2,1}^{H,out}$	802.750117011701		
$b_{2,4,2}^{H,out}$	1460.6800000000012		
$b_{3,5,1}^{H,out}$	628.835654872749		
$b_{4,3,2}^{H,out}$	1876.1399999999996		
$b_{5,1,1}^{H,out}$	85.15696924229705		

3 *RecLMTD* Tangent Points

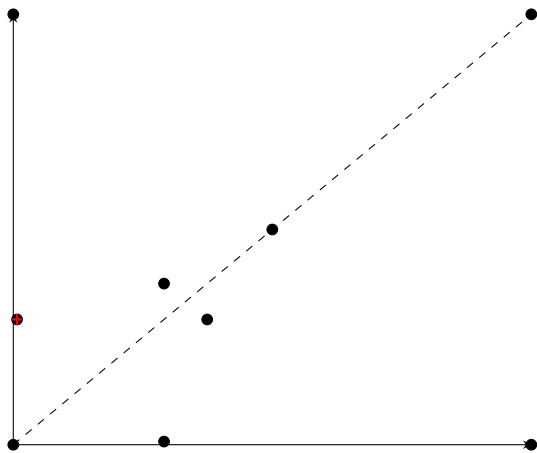
- (1, 1, 2)
 - Value: 0.036098
 - Error: 0.000000964164276
 - Relative error: 0.000026708653087
- (2, 2, 1)
 - Value: 0.021686
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000015
- (2, 4, 2)
 - Value: 0.036156
 - Error: 0.000000038617563
 - Relative error: 0.000001068072592
- (3, 5, 1)
 - Value: 0.039171
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000001
- (4, 3, 2)
 - Value: 0.013066
 - Error: 0.000000985435591
 - Relative error: 0.000075412743761
- (5, 1, 1)
 - Value: 0.021740
 - Error: 0.000440298415720
 - Relative error: 0.019850636878759
- (5, 4, 2)
 - Value: 0.039761
 - Error: 0.0000000000000003
 - Relative error: 0.0000000000000064
- $cu, 3$
 - Value: 0.016938
 - Error: 0.000000923579561
 - Relative error: 0.000054525551807
- $cu, 5$
 - Value: 0.016454
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000007



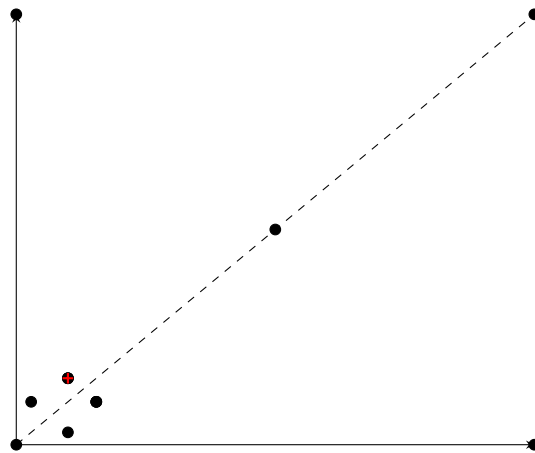
(a) RecLMTD - (1, 1, 2)



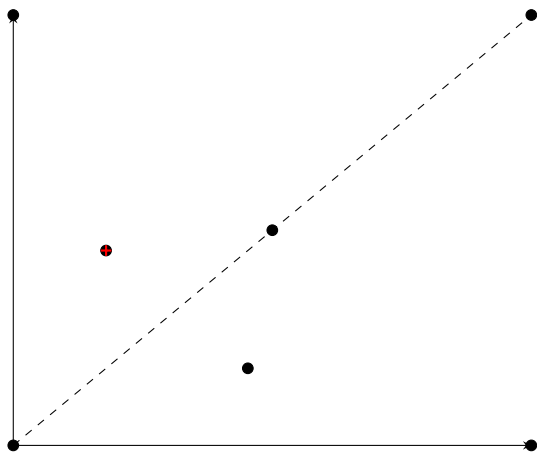
(b) RecLMTD - (2, 2, 1)



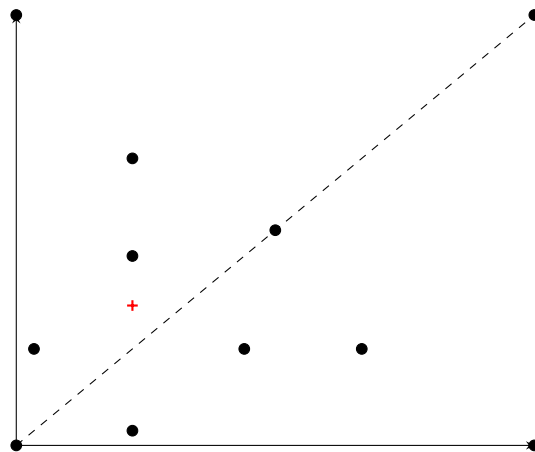
(c) RecLMTD - (2, 4, 2)



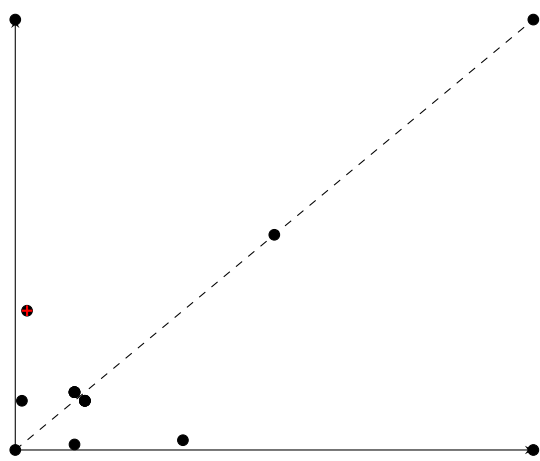
(d) RecLMTD - (3, 5, 1)



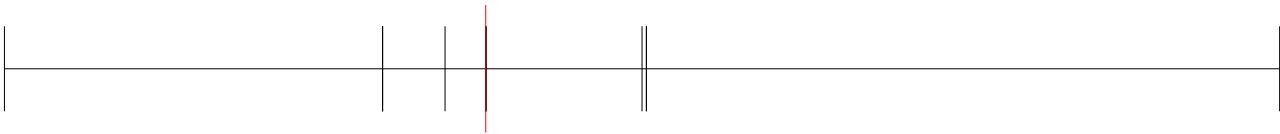
(e) RecLMTD - (4, 3, 2)



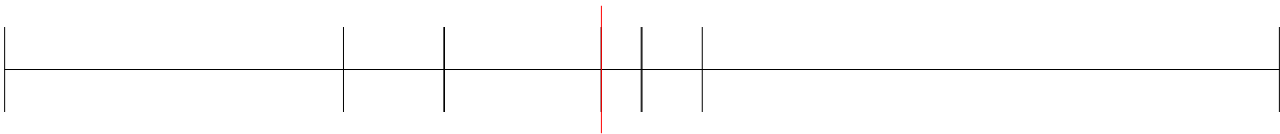
(f) RecLMTD - (5, 1, 1)



(g) *RecLMTD* - (5, 4, 2)



(a) *RecLMTD* - *cu*, 3



(b) *RecLMTD* - *cu*, 5

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 2)
 - Value: 1408.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 1449.960117011701
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 4, 2)
 - Value: 1991.1299999999924
 - Absolute error: 6.821210263296962e-12
 - Relative error: 3.4257985482098163e-15
- (3, 5, 1)
 - Value: 2173.1256548727492
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 3415.8599999999997
 - Absolute error: 4.547473508864641e-13
 - Relative error: -1.3312821687260722e-16
- (5, 1, 1)
 - Value: 258.1969692422834
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 2)
 - Value: 1690.915973051231
 - Absolute error: 1.7280399333685637e-11
 - Relative error: 1.021954940936752e-14

4.2 $b^{H,\text{out}}$

- (1, 1, 2)
 - Value: 821.0400000000002
 - Absolute error: 1.1368683772161603e-13
 - Relative error: 1.3846686851020172e-16
- (2, 2, 1)

- Value: 802.750117011701
 - Absolute error: 5.404958903552597
 - Relative error: -0.0066880219708220735
- (2, 4, 2)
 - Value: 1460.6800000000012
 - Absolute error: 1.1368683772161603e-12
 - Relative error: 7.783144680670375e-16
- (3, 5, 1)
 - Value: 628.835654872749
 - Absolute error: 6.821210263296962e-13
 - Relative error: -1.0847365620000176e-15
- (4, 3, 2)
 - Value: 1876.1399999999996
 - Absolute error: 4.547473508864641e-13
 - Relative error: -2.4238455066597593e-16
- (5, 1, 1)
 - Value: 85.15696924229705
 - Absolute error: 4.547473508864641e-12
 - Relative error: -5.34010727404527e-14
- (5, 4, 2)
 - Value: 586.5159730512163
 - Absolute error: 3.410605131648481e-13
 - Relative error: -5.815025145701624e-16

4.3 $b^{C,in}$

- (1, 1, 2)
 - Value: 352.17600000000001
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 705.16
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 4, 2)
 - Value: 461.40730158728866
 - Absolute error: 0.0
 - Relative error: 0.0

- (3, 5, 1)
 - Value: 1296.87
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 317.52
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 1, 1)
 - Value: 1042.9600000000273
 - Absolute error: 2.8194335754960775e-11
 - Relative error: 2.7032998154254044e-14
- (5, 4, 2)
 - Value: 960.6526984127114
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 2)
 - Value: 939.1360000000002
 - Absolute error: 1.1368683772161603e-13
 - Relative error: -1.2105471169416996e-16
- (2, 2, 1)
 - Value: 1352.37
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 4, 2)
 - Value: 991.8573015872737
 - Absolute error: 189.14898959572918
 - Relative error: 0.2356385087460283
- (3, 5, 1)
 - Value: 2841.1600000000003
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 1857.24
 - Absolute error: 0.0

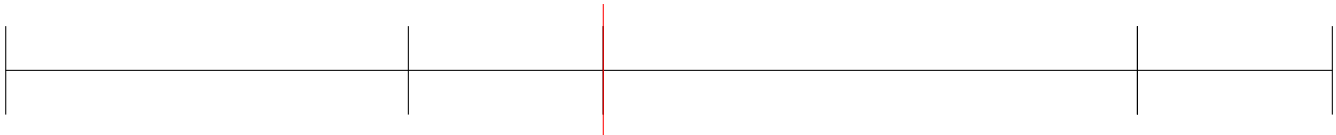
- Relative error: 0.0
- (5, 1, 1)
 - Value: 1216.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 4, 2)
 - Value: 2065.0526984127264
 - Absolute error: 0.0
 - Relative error: 0.0



(a) $t^{(H)} - (1, 2)$



(a) $t^{(H)} - (2, 1)$



(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (3, 1)$



(a) $t^{(H)} - (4, 2)$



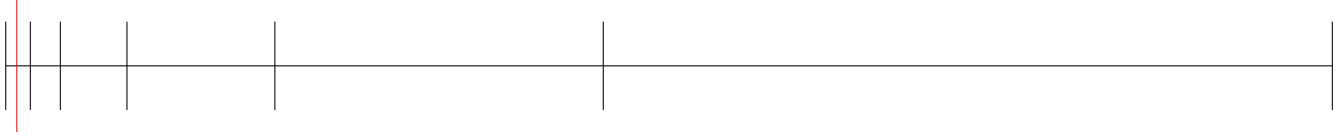
(a) $t^{(H)} - (5, 1)$



(a) $t^{(H)} - (5, 2)$



(a) $t^{(H)} - (1, 1, 2)$



(a) $t^{(H)} - (2, 2, 1)$



(a) $t^{(H)} - (2, 4, 2)$



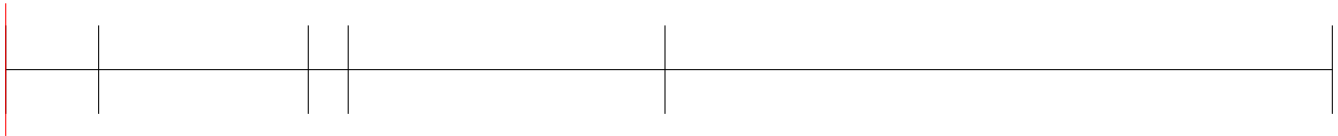
(a) $t^{(H)} - (3, 5, 1)$



(a) $t^{(H)} - (4, 3, 2)$



(a) $t^{(H)} - (5, 1, 1)$



(a) $t^{(H)} - (5, 4, 2)$



(a) $t^{(C)} - (1, 3)$



(a) $t^{(C)} - (2, 2)$



(a) $t^{(C)} - (4, 3)$



(a) $t^{(C)} - (5, 2)$



(a) $t^{(C)} - (3, 3)$



(a) $t^{(C)} - (1, 2)$



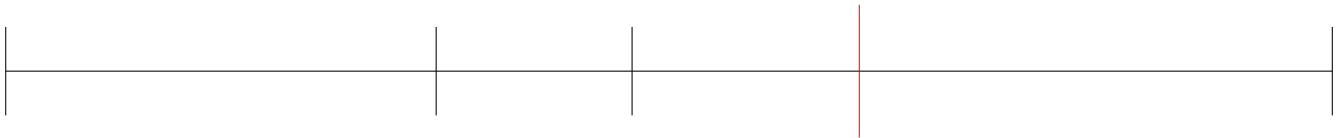
(a) $t^{(C)} - (4, 3)$



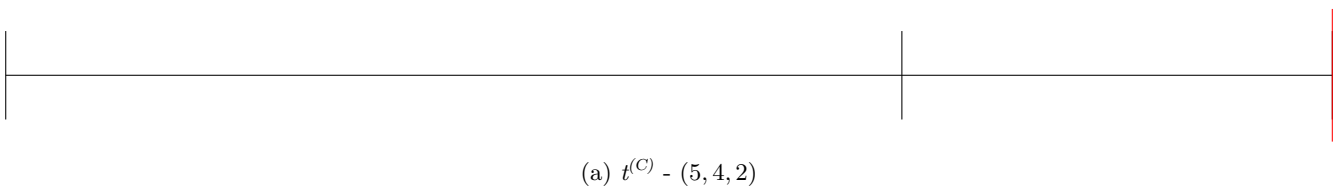
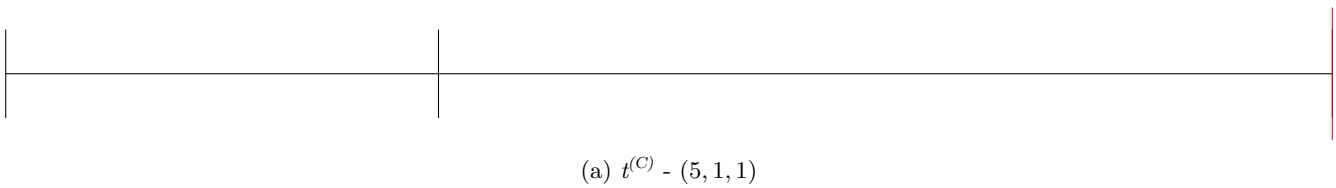
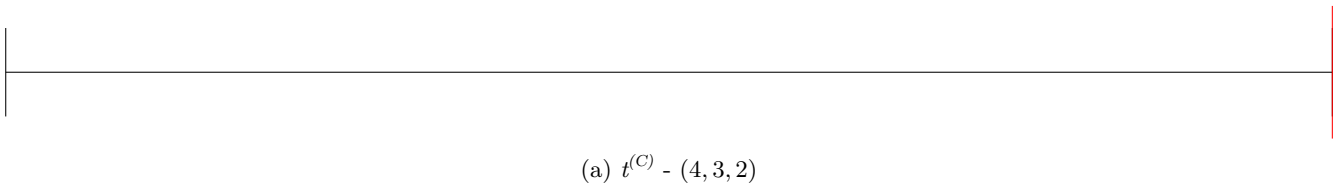
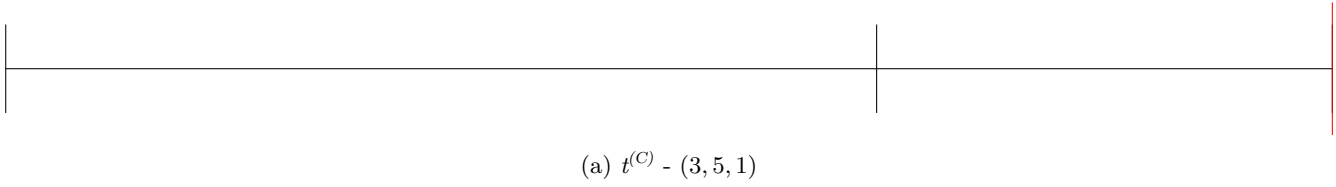
(a) $t^{(C)} - (1, 1, 2)$



(a) $t^{(C)} - (2, 2, 1)$



(a) $t^{(C)} - (2, 4, 2)$



5 Area Breakpoints

5.1 A

- $(1, 1, 2)$
 - Value: 24.927400
 - Error: 0.000002301803917
 - Relative error: 0.000000092340303
- $(2, 2, 1)$
 - Value: 16.512573
 - Error: 0.000000000000661
 - Relative error: 0.00000000000040
- $(2, 4, 2)$
 - Value: 22.563640
 - Error: 0.00000000000014
 - Relative error: -0.00000000000001
- $(3, 5, 1)$
 - Value: 71.165974
 - Error: 0.00000000000199
 - Relative error: 0.00000000000003
- $(4, 3, 2)$
 - Value: 23.668655
 - Error: 0.00000000000000
 - Relative error: 0.00000000000000
- $(5, 1, 1)$
 - Value: 4.425808
 - Error: 0.00000000001299
 - Relative error: 0.00000000000294
- $(5, 4, 2)$
 - Value: 51.661546
 - Error: 0.00000000000412
 - Relative error: 0.00000000000008

5.2 A_{CU}

- $cu, 3$
 - Value: 16.738084
 - Error: 0.00000000000032
 - Relative error: -0.00000000000002
- $cu, 5$

- Value: 20.944873
- Error: 0.000000000000064
- Relative error: 0.000000000000003



(a) $A^\beta - (1, 1, 2)$



(a) $A^\beta - (2, 2, 1)$



(a) $A^\beta - (2, 4, 2)$



(a) $A^\beta - (3, 5, 1)$



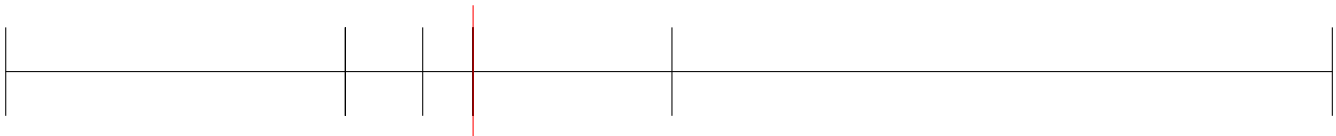
(a) $A^\beta - (4, 3, 2)$



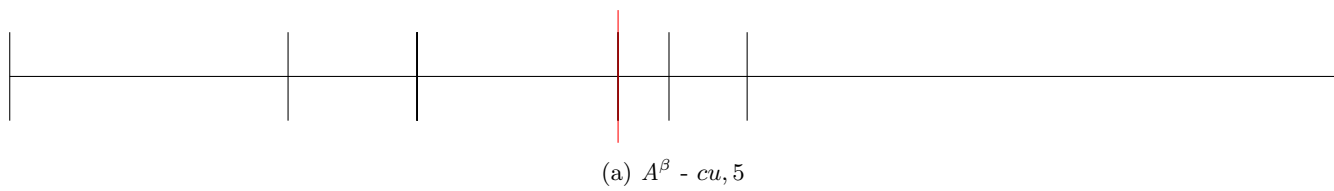
(a) $A^\beta - (5, 1, 1)$



(a) $A^\beta - (5, 4, 2)$



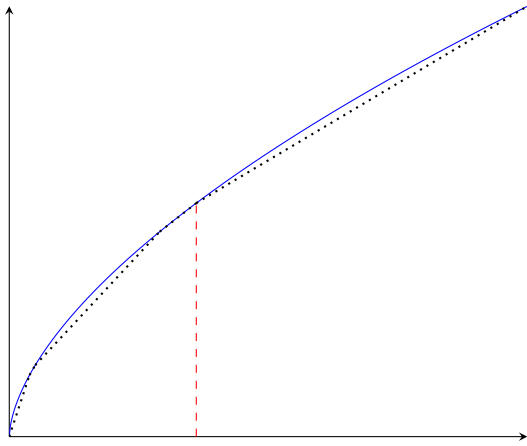
(a) $A^\beta - cu, 3$



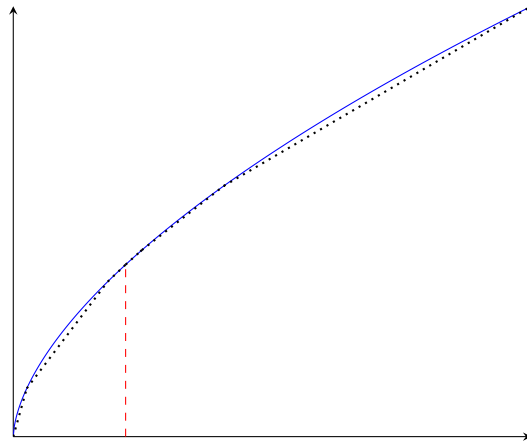
6 A^β Breakpoints

- (1, 1, 2)
 - q : 24.927400
 - $\hat{q}^\beta$: 6.886621
 - q^β : 6.886621
 - Breakpoints: [0, 2.7005882352941173, 3.460657649345137, 20.05363555217427, 24.927400386350968, 24.928068483287923, 24.928068483288385, 69.05411764705883]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 2, 1)
 - q : 16.512573
 - $\hat{q}^\beta$: 5.378843
 - q^β : 5.378843
 - Breakpoints: [0, 1.9849079917037988, 13.879417919656465, 16.512573095835464, 18.676744629514324, 18.890814020083003, 30.93278282491032, 30.932782824910646, 30.93278282491164, 76.14235294117645]
 - Error: 0.0000000000000034
 - Relative error: 0.0000000000000006
- (2, 4, 2)
 - q : 22.563640
 - $\hat{q}^\beta$: 4.590400
 - q^β : 6.487024
 - Breakpoints: [0, 2.5223896397191172, 3.743604220332404, 138.54823529411763]
 - Error: 1.896624095116719
 - Relative error: 0.292371995699319
- (3, 5, 1)
 - q : 71.165974
 - $\hat{q}^\beta$: 12.923003
 - q^β : 12.923003
 - Breakpoints: [0, 30.750913778350093, 44.57376487924321, 71.16597394333036, 71.16597394333074, 181.68117647058827]
 - Error: 0.0000000000000002
 - Relative error: 0.0000000000000000
- (4, 3, 2)
 - q : 23.668655
 - $\hat{q}^\beta$: 6.675779
 - q^β : 6.675813
 - Breakpoints: [0, 19.92063386984238, 23.66863193022334, 23.668654733871367, 23.668654734065058, 23.668855770059167, 23.668855770059224, 181.14352941176475]
 - Error: 0.000034135787302

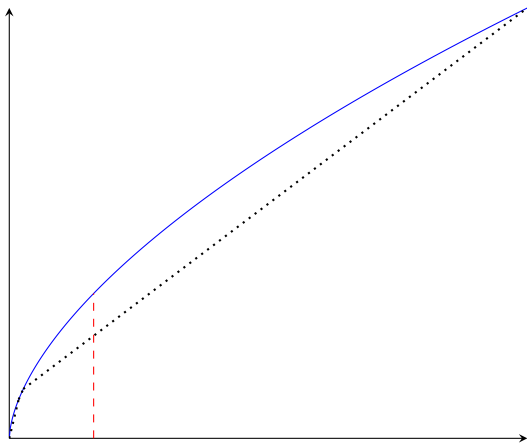
- Relative error: 0.000005113353046
- (5, 1, 1)
 - q : 4.425808
 - $\hat{q}^\beta$: 2.128002
 - q^β : 2.441156
 - Breakpoints: [0, 1.4656333375682882, 13.141872711246277, 16.623773924385056, 89.41176470588236]
 - Error: 0.313153733272757
 - Relative error: 0.128280924514211
- (5, 4, 2)
 - q : 51.661546
 - $\hat{q}^\beta$: 10.660837
 - q^β : 10.663516
 - Breakpoints: [0, 11.133625686778567, 51.43476571725509, 82.7196877037192, 82.71968770371944, 82.71968770371991, 106.27900213658211, 192.3352941176471]
 - Error: 0.002678980346593
 - Relative error: 0.000251228606524
- $cu, 3$
 - q : 16.738084
 - $\hat{q}^\beta$: 5.422798
 - q^β : 5.422798
 - Breakpoints: [0, 7.499856372317694, 10.320104112088831, 13.6435791010403, 13.643579101040302, 14.376341772690381, 16.738084386188994, 16.73892021814917, 16.73899708924428, 16.738997089244634, 16.738997089244645, 18.852085704732907, 20.61398815516275, 128.44338870924773]
 - Error: 0.0000000000000016
 - Relative error: -0.0000000000000003
- $cu, 5$
 - q : 20.944873
 - $\hat{q}^\beta$: 6.203641
 - q^β : 6.203641
 - Breakpoints: [0, 6.717165864481, 9.832103793393394, 14.682070969056184, 16.23492858720698, 16.234975184591587, 16.235135805353295, 20.944873037188774, 20.94487303718885, 20.944873037188852, 21.50633939847323, 22.410514202757604, 127.1036185995295]
 - Error: 0.0000000000000002
 - Relative error: -0.0000000000000000



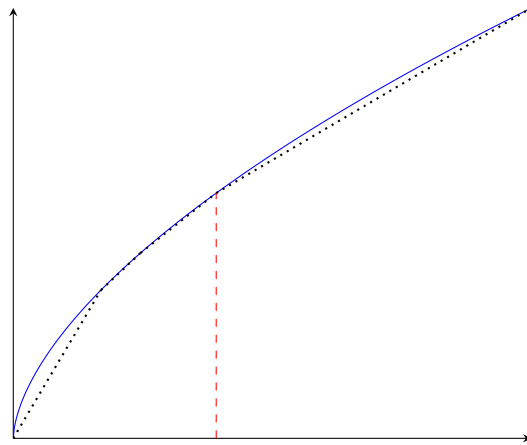
(a) $q^\beta - (1, 1, 2)$



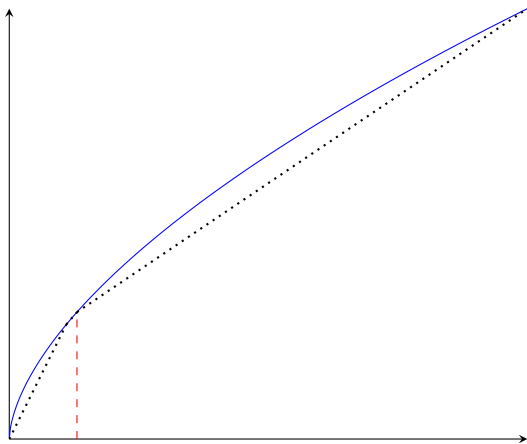
(b) $q^\beta - (2, 2, 1)$



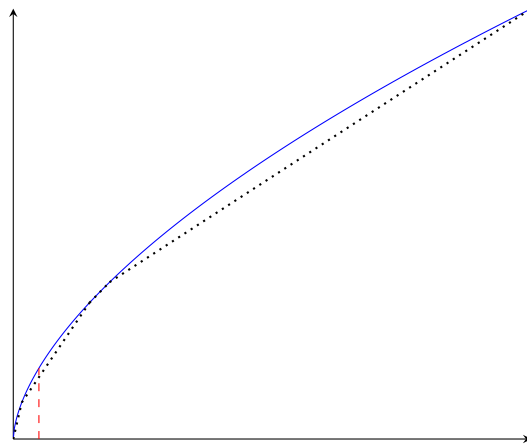
(c) $q^\beta - (2, 4, 2)$



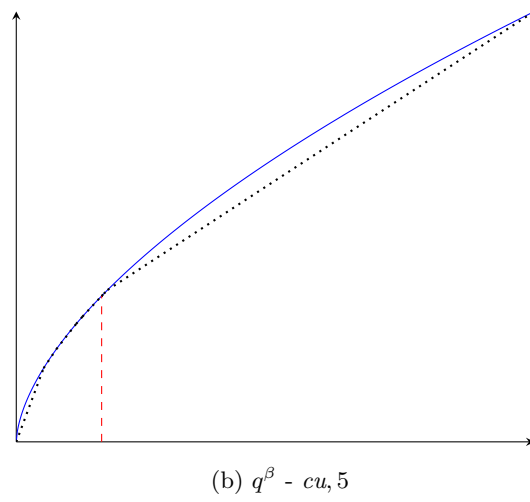
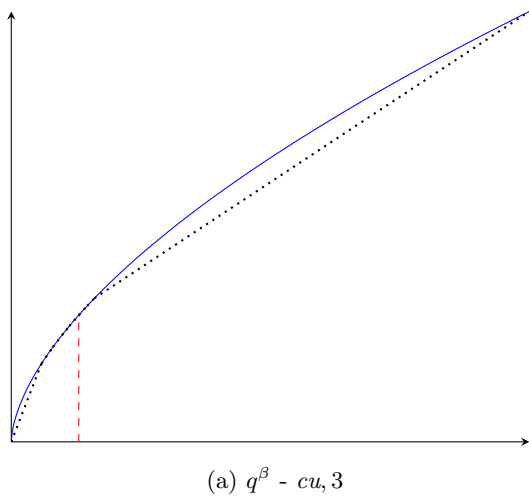
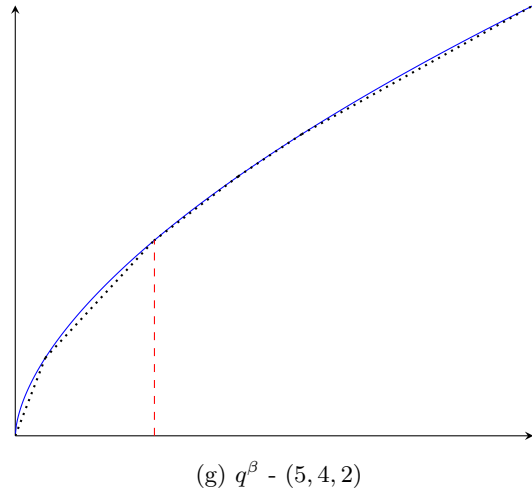
(d) $q^\beta - (3, 5, 1)$



(e) $q^\beta - (4, 3, 2)$



(f) $q^\beta - (5, 1, 1)$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	-0.0	$z_{4,5,1}$	-0.0	$A_{2,5,1}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{4,5,2}$	-0.0	$A_{2,5,2}^\beta$	0.0
$z_{1,2,2}$	-0.0	$z_{5,1,2}$	-0.0	$A_{3,1,1}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,2,1}$	0.0	$A_{3,1,2}^\beta$	0.0
$z_{1,3,2}$	-0.0	$z_{5,2,2}$	-0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,3,1}$	0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,4,2}$	-0.0	$z_{5,3,2}$	0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,4,1}$	-0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,2}$	-0.0	$z_{5,5,1}$	0.0	$A_{3,4,1}^\beta$	0.0
$z_{2,1,1}$	-0.0	$z_{5,5,2}$	-0.0	$A_{3,4,2}^\beta$	0.0
$z_{2,1,2}$	-0.0	$z_{cu,1}$	-0.0	$A_{3,5,2}^\beta$	0.0
$z_{2,2,2}$	-0.0	$z_{cu,2}$	-0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,3,1}$	0.0	$z_{cu,4}$	0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,3,2}$	-0.0	$z_{hu,1}$	-0.0	$A_{4,2,1}^\beta$	0.0
$z_{2,4,1}$	-0.0	$z_{hu,2}$	-0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,5,1}$	-0.0	$z_{hu,3}$	-0.0	$A_{4,3,1}^\beta$	0.0
$z_{2,5,2}$	-0.0	$z_{hu,4}$	-0.0	$A_{4,4,1}^\beta$	0.0
$z_{3,1,1}$	-0.0	$z_{hu,5}$	-0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,2}$	-0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,5,1}^\beta$	0.0
$z_{3,2,1}$	-0.0	$A_{1,2,1}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,2,2}$	-0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,1,2}^\beta$	0.0
$z_{3,3,1}$	-0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,3,2}$	-0.0	$A_{1,3,2}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,4,1}$	-0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,4,2}$	-0.0	$A_{1,4,2}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{3,5,2}$	-0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,4,1}^\beta$	0.0
$z_{4,1,1}$	-0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,1,2}$	-0.0	$A_{2,1,1}^\beta$	0.0	$A_{5,5,2}^\beta$	0.0
$z_{4,2,1}$	-0.0	$A_{2,1,2}^\beta$	0.0	$A_{cu,1}^\beta$	0.0
$z_{4,2,2}$	-0.0	$A_{2,2,2}^\beta$	0.0	$A_{cu,2}^\beta$	0.0
$z_{4,3,1}$	-0.0	$A_{2,3,1}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,4,1}$	-0.0	$A_{2,3,2}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,4,2}$	-0.0	$A_{2,4,1}^\beta$	0.0	$A_{hu,2}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,3}^{\beta}$	0.0	$A_{4,3,1}$	0.0	$dt_{1,5,3}$	66.7
$A_{hu,4}^{\beta}$	0.0	$A_{4,4,1}$	0.0	$dt_{2,1,1}$	111.1
$A_{hu,5}^{\beta}$	0.0	$A_{4,4,2}$	0.0	$dt_{2,1,2}$	72.81087388281803
$A_{1,1,1}$	0.0	$A_{4,5,1}$	0.0	$dt_{2,1,3}$	100.0
$A_{1,2,1}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,2,3}$	106.1
$A_{1,2,2}$	0.0	$A_{5,1,2}$	0.0	$dt_{2,3,1}$	111.1
$A_{1,3,1}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,3,2}$	10.0
$A_{1,3,2}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,3,3}$	183.3
$A_{1,4,1}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,4,1}$	111.1
$A_{1,4,2}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,5,1}$	111.1
$A_{1,5,1}$	0.0	$A_{5,4,1}$	0.0	$dt_{2,5,2}$	52.5049748880142
$A_{1,5,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{2,5,3}$	111.1
$A_{2,1,1}$	0.0	$A_{5,5,2}$	0.0	$dt_{3,1,1}$	161.1
$A_{2,1,2}$	0.0	$A_{cu,1}$	0.0	$dt_{3,1,2}$	10.0
$A_{2,2,2}$	0.0	$A_{cu,2}$	0.0	$dt_{3,1,3}$	10.0
$A_{2,3,1}$	0.0	$A_{cu,4}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,3,2}$	0.0	$A_{hu,1}$	0.0	$dt_{3,2,2}$	111.1
$A_{2,4,1}$	0.0	$A_{hu,2}$	0.0	$dt_{3,2,3}$	111.1
$A_{2,5,1}$	0.0	$A_{hu,3}$	0.0	$dt_{3,3,1}$	161.1
$A_{2,5,2}$	0.0	$A_{hu,4}$	0.0	$dt_{3,3,2}$	10.0
$A_{3,1,1}$	0.0	$A_{hu,5}$	0.0	$dt_{3,3,3}$	188.9
$A_{3,1,2}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,4,1}$	144.5
$A_{3,2,1}$	0.0	$dt_{1,2,1}$	17.15006133296988	$dt_{3,4,2}$	10.0
$A_{3,2,2}$	0.0	$dt_{1,2,2}$	17.15006133296988	$dt_{3,4,3}$	144.5
$A_{3,3,1}$	0.0	$dt_{1,2,3}$	17.15006133296988	$dt_{3,5,3}$	133.4
$A_{3,3,2}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,1,1}$	122.2
$A_{3,4,1}$	0.0	$dt_{1,3,2}$	10.0	$dt_{4,1,2}$	144.96842105263215
$A_{3,4,2}$	0.0	$dt_{1,3,3}$	122.2	$dt_{4,1,3}$	88.9
$A_{3,5,2}$	0.0	$dt_{1,4,1}$	10.0	$dt_{4,2,1}$	122.2
$A_{4,1,1}$	0.0	$dt_{1,4,2}$	66.7	$dt_{4,2,2}$	155.5
$A_{4,1,2}$	0.0	$dt_{1,4,3}$	77.8	$dt_{4,2,3}$	33.3
$A_{4,2,1}$	0.0	$dt_{1,5,1}$	66.7	$dt_{4,3,1}$	122.2
$A_{4,2,2}$	0.0	$dt_{1,5,2}$	66.7	$dt_{4,4,1}$	122.2

Variable	Value	Variable	Value	Variable	Value
$dt_{4,4,2}$	122.2	$f_{2,1,1}^H$	0.0	$f_{5,5,2}^H$	0.0
$dt_{4,4,3}$	66.7	$f_{2,1,2}^H$	0.0	$f_{1,1,1}^C$	0.0
$dt_{4,5,1}$	122.2	$f_{2,2,2}^H$	0.0	$f_{1,2,1}^C$	0.0
$dt_{4,5,2}$	177.8	$f_{2,3,1}^H$	0.0	$f_{1,2,2}^C$	0.0
$dt_{4,5,3}$	55.6	$f_{2,3,2}^H$	0.0	$f_{1,3,1}^C$	8.4
$dt_{5,1,3}$	10.0	$f_{2,4,1}^H$	0.0	$f_{1,3,2}^C$	0.0
$dt_{5,2,1}$	74.66004781348396	$f_{2,5,1}^H$	4.774527452745216	$f_{1,4,1}^C$	0.0
$dt_{5,2,2}$	74.66004781348396	$f_{2,5,2}^H$	0.0	$f_{1,4,2}^C$	0.0
$dt_{5,2,3}$	10.0	$f_{3,1,1}^H$	0.0	$f_{1,5,1}^C$	0.0
$dt_{5,3,1}$	133.3	$f_{3,1,2}^H$	13.499755513902487	$f_{1,5,2}^C$	13.9
$dt_{5,3,2}$	10.0	$f_{3,2,1}^H$	5.21409062693979	$f_{2,1,1}^C$	0.0
$dt_{5,3,3}$	54.4	$f_{3,2,2}^H$	0.0	$f_{2,1,2}^C$	0.0
$dt_{5,4,1}$	116.7	$f_{3,3,1}^H$	0.0	$f_{2,2,2}^C$	6.1
$dt_{5,5,1}$	105.6	$f_{3,3,2}^H$	0.0	$f_{2,3,1}^C$	0.0
$dt_{5,5,2}$	105.6	$f_{3,4,1}^H$	0.0	$f_{2,3,2}^C$	0.0
$dt_{5,5,3}$	105.6	$f_{3,4,2}^H$	1.300244486097502	$f_{2,4,1}^C$	0.0
$dt_{cu,1}$	53.3	$f_{3,5,2}^H$	0.0	$f_{2,5,1}^C$	0.0
$dt_{cu,2}$	97.8	$f_{4,1,1}^H$	12.6	$f_{2,5,2}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,1,2}^H$	0.0	$f_{3,1,1}^C$	0.0
$dt_{hu,1}$	80.0	$f_{4,2,1}^H$	0.0	$f_{3,1,2}^C$	0.0
$dt_{hu,2}$	18.3	$f_{4,2,2}^H$	0.0	$f_{3,2,1}^C$	0.0
$dt_{hu,3}$	18.9	$f_{4,3,1}^H$	0.0	$f_{3,2,2}^C$	0.0
$dt_{hu,4}$	63.3	$f_{4,4,1}^H$	0.0	$f_{3,3,1}^C$	0.0
$dt_{hu,5}$	35.6	$f_{4,4,2}^H$	0.0	$f_{3,3,2}^C$	0.0
$f_{1,1,1}^H$	8.8	$f_{4,5,1}^H$	0.0	$f_{3,4,1}^C$	0.0
$f_{1,2,1}^H$	0.0	$f_{4,5,2}^H$	0.0	$f_{3,4,2}^C$	0.0
$f_{1,2,2}^H$	0.0	$f_{5,1,2}^H$	0.0	$f_{3,5,2}^C$	0.0
$f_{1,3,1}^H$	0.0	$f_{5,2,1}^H$	0.0	$f_{4,1,1}^C$	0.0
$f_{1,3,2}^H$	0.0	$f_{5,2,2}^H$	8.759207727877815	$f_{4,1,2}^C$	1.7303999999999817
$f_{1,4,1}^H$	0.0	$f_{5,3,1}^H$	16.401875468867356	$f_{4,2,1}^C$	0.0
$f_{1,4,2}^H$	0.0	$f_{5,3,2}^H$	0.0	$f_{4,2,2}^C$	0.0
$f_{1,5,1}^H$	0.0	$f_{5,4,1}^H$	0.0	$f_{4,3,1}^C$	0.0
$f_{1,5,2}^H$	0.0	$f_{5,5,1}^H$	0.0	$f_{4,4,1}^C$	17.3

Variable	Value	Variable	Value	Variable	Value
$f_{4,4,2}^C$	0.0	$\text{RecLMTD}_{3,3,2}$	0.026450800778310225	$q_{1,3,2}$	0.0
$f_{4,5,1}^C$	0.0	$\text{RecLMTD}_{3,4,1}$		$q_{1,4,1}$	0.0
$f_{4,5,2}^C$	0.0	$\text{RecLMTD}_{3,4,2}$		$q_{1,4,2}$	0.0
$f_{5,1,2}^C$	0.0	$\text{RecLMTD}_{3,5,2}$		$q_{1,5,1}$	0.0
$f_{5,2,1}^C$	0.0	$\text{RecLMTD}_{4,1,1}$	0.020350921041481694	$q_{1,5,2}$	0.0
$f_{5,2,2}^C$	0.0	$\text{RecLMTD}_{4,1,2}$	0.02150212335787724	$q_{2,1,1}$	0.0
$f_{5,3,1}^C$	0.0	$\text{RecLMTD}_{4,2,1}$	0.007119446655845005	$q_{2,1,2}$	0.0
$f_{5,3,2}^C$	0.0	$\text{RecLMTD}_{4,2,2}$	0.029297885186065308	$q_{2,2,2}$	0.0
$f_{5,4,1}^C$	0.0	$\text{RecLMTD}_{4,3,1}$		$q_{2,3,1}$	0.0
$f_{5,5,1}^C$	0.0	$\text{RecLMTD}_{4,4,1}$	0.02230651890829312	$q_{2,3,2}$	0.0
$f_{5,5,2}^C$	0.0	$\text{RecLMTD}_{4,4,2}$	0.024030191698075007	$q_{2,4,1}$	0.0
$\text{RecLMTD}_{1,1,1}$	0.1	$\text{RecLMTD}_{4,5,1}$	0.006503686476760015	$q_{2,5,1}$	0.0
$\text{RecLMTD}_{1,2,1}$	0.1	$\text{RecLMTD}_{4,5,2}$	0.02554670259920889	$q_{2,5,2}$	0.0
$\text{RecLMTD}_{1,2,2}$	0.1	$\text{RecLMTD}_{5,1,2}$		$q_{3,1,1}$	0.0
$\text{RecLMTD}_{1,3,1}$	0.1	$\text{RecLMTD}_{5,2,1}$		$q_{3,1,2}$	0.0
$\text{RecLMTD}_{1,3,2}$	0.1	$\text{RecLMTD}_{5,2,2}$		$q_{3,2,1}$	0.0
$\text{RecLMTD}_{1,4,1}$	0.1	$\text{RecLMTD}_{5,3,1}$		$q_{3,2,2}$	0.0
$\text{RecLMTD}_{1,4,2}$	0.01377039538464575	$\text{RecLMTD}_{5,3,2}$	0.02700088599025016	$q_{3,3,1}$	0.0
$\text{RecLMTD}_{1,5,1}$	0.1	$\text{RecLMTD}_{5,4,1}$		$q_{3,3,2}$	0.0
$\text{RecLMTD}_{1,5,2}$	0.1	$\text{RecLMTD}_{5,5,1}$		$q_{3,4,1}$	0.0
$\text{RecLMTD}_{2,1,1}$	0.1	$\text{RecLMTD}_{5,5,2}$	0.009469696969696968	$q_{3,4,2}$	0.0
$\text{RecLMTD}_{2,1,2}$	0.1	$\text{RecLMTD}_{cu,1}$	0.016108908592626727	$q_{3,5,2}$	0.0
$\text{RecLMTD}_{2,2,2}$	0.1	$\text{RecLMTD}_{cu,2}$	0.009423141115768875	$q_{4,1,1}$	0.0
$\text{RecLMTD}_{2,3,1}$	0.049286610717187346	$\text{RecLMTD}_{cu,4}$	0.008535688158765427	$q_{4,1,2}$	0.0
$\text{RecLMTD}_{2,3,2}$	0.020926522199679467	$\text{RecLMTD}_{hu,1}$	0.012387051800747543	$q_{4,2,1}$	0.0
$\text{RecLMTD}_{2,4,1}$	0.1	$\text{RecLMTD}_{hu,2}$	0.045618305265862594	$q_{4,2,2}$	0.0
$\text{RecLMTD}_{2,5,1}$	0.1	$\text{RecLMTD}_{hu,3}$	0.04305097282506776	$q_{4,3,1}$	0.0
$\text{RecLMTD}_{2,5,2}$	0.1	$\text{RecLMTD}_{hu,4}$	0.015434474530129737	$q_{4,4,1}$	0.0
$\text{RecLMTD}_{3,1,1}$	0.1	$\text{RecLMTD}_{hu,5}$	0.024141568940966463	$q_{4,4,2}$	0.0
$\text{RecLMTD}_{3,1,2}$	0.1	$q_{1,1,1}$		$q_{4,5,1}$	0.0
$\text{RecLMTD}_{3,2,1}$	0.1	$q_{1,2,1}$		$q_{4,5,2}$	0.0
$\text{RecLMTD}_{3,2,2}$	0.1	$q_{1,2,2}$		$q_{5,1,2}$	0.0
$\text{RecLMTD}_{3,3,1}$	0.1	$q_{1,3,1}$		$q_{5,2,1}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{5,2,2}$	0.0	$t_{3,2,1}^H$	226.7	$t_{2,1,1}^C$	141.65704735640725
$q_{5,3,1}$	0.0	$t_{3,2,2}^H$	65.6	$t_{2,1,2}^C$	129.79605263157896
$q_{5,3,2}$	0.0	$t_{3,3,1}^H$	226.7	$t_{2,2,2}^C$	115.6
$q_{5,4,1}$	0.0	$t_{3,3,2}^H$	65.6	$t_{2,3,1}^C$	221.1
$q_{5,5,1}$	0.0	$t_{3,4,1}^H$	226.7	$t_{2,3,2}^C$	37.8
$q_{5,5,2}$	0.0	$t_{3,4,2}^H$	116.34846317269022	$t_{2,4,1}^C$	82.2
$q_{cu,1}$	0.0	$t_{3,5,2}^H$	65.6	$t_{2,5,1}^C$	106.45135285116358
$q_{cu,2}$	0.0	$t_{4,1,1}^H$	271.1	$t_{2,5,2}^C$	93.3
$q_{cu,4}$	0.0	$t_{4,1,2}^H$	271.1	$t_{3,1,1}^C$	137.6462144488957
$q_{hu,1}$	0.0	$t_{4,2,1}^H$	271.1	$t_{3,1,2}^C$	90.20394736842073
$q_{hu,2}$	0.0	$t_{4,2,2}^H$	271.1	$t_{3,2,1}^C$	115.6
$q_{hu,3}$	0.0	$t_{4,3,1}^H$	271.1	$t_{3,2,2}^C$	115.6
$q_{hu,4}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,3,1}^C$	221.1
$q_{hu,5}$	0.0	$t_{4,4,2}^H$	271.1	$t_{3,3,2}^C$	37.8
$t_{1,1,1}^H$	160.0	$t_{4,5,1}^H$	271.1	$t_{3,4,1}^C$	176.7
$t_{1,2,1}^H$	160.0	$t_{4,5,2}^H$	271.1	$t_{3,4,2}^C$	82.2
$t_{1,2,2}^H$	93.3	$t_{5,1,2}^H$	91.18281635356577	$t_{3,5,2}^C$	93.3
$t_{1,3,1}^H$	160.0	$t_{5,2,1}^H$	198.9	$t_{4,1,1}^C$	70.26757367219665
$t_{1,3,2}^H$	160.0	$t_{5,2,2}^H$	148.52371394662657	$t_{4,1,2}^C$	60.0
$t_{1,4,1}^H$	160.0	$t_{5,3,1}^H$	198.9	$t_{4,2,1}^C$	221.7
$t_{1,4,2}^H$	160.0	$t_{5,3,2}^H$	65.6	$t_{4,2,2}^C$	115.6
$t_{1,5,1}^H$	160.0	$t_{5,4,1}^H$	198.9	$t_{4,3,1}^C$	221.1
$t_{1,5,2}^H$	160.0	$t_{5,5,1}^H$	198.9	$t_{4,4,1}^C$	176.7
$t_{2,1,1}^H$	139.0711116115888	$t_{5,5,2}^H$	65.60191887928448	$t_{4,4,2}^C$	82.2
$t_{2,1,2}^H$	137.8	$t_{1,1,1}^C$	100.47152104061766	$t_{4,5,1}^C$	185.0902773345454
$t_{2,2,2}^H$	137.8	$t_{1,2,1}^C$	115.6	$t_{4,5,2}^C$	93.3
$t_{2,3,1}^H$	248.9	$t_{1,2,2}^C$	115.6	$t_{5,1,2}^C$	90.20394736842104
$t_{2,3,2}^H$	137.8	$t_{1,3,1}^C$	221.1	$t_{5,2,1}^C$	115.6
$t_{2,4,1}^H$	198.8575471698113	$t_{1,3,2}^C$	107.67619047619047	$t_{5,2,2}^C$	115.6
$t_{2,5,1}^H$	248.9	$t_{1,4,1}^C$	176.7	$t_{5,3,1}^C$	221.1
$t_{2,5,2}^H$	137.8	$t_{1,4,2}^C$	129.3704699676358	$t_{5,3,2}^C$	37.8
$t_{3,1,1}^H$	105.65126083492889	$t_{1,5,1}^C$	168.01641176300524	$t_{5,4,1}^C$	176.7
$t_{3,1,2}^H$	109.22992778744405	$t_{1,5,2}^C$	93.3	$t_{5,5,1}^C$	168.01641176300535

Variable	Value	Variable	Value	Variable	Value
$t_{5,5,2}^C$	124.01079136690645	$b_{4,4,2}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	0.0
$b_{1,1,1}^{H,in}$	1408.0	$b_{4,5,1}^{H,in}$	0.0	$b_{3,4,1}^{H,out}$	0.0
$b_{1,2,1}^{H,in}$	0.0	$b_{4,5,2}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	279.5631070338243
$b_{1,2,2}^{H,in}$	0.0	$b_{5,1,2}^{H,in}$	0.0	$b_{3,5,2}^{H,out}$	0.0
$b_{1,3,1}^{H,in}$	0.0	$b_{5,2,1}^{H,in}$	0.0	$b_{4,1,1}^{H,out}$	3415.86
$b_{1,3,2}^{H,in}$	0.0	$b_{5,2,2}^{H,in}$	1656.5740269487962	$b_{4,1,2}^{H,out}$	0.0
$b_{1,4,1}^{H,in}$	0.0	$b_{5,3,1}^{H,in}$	3262.333030757717	$b_{4,2,1}^{H,out}$	0.0
$b_{1,4,2}^{H,in}$	0.0	$b_{5,3,2}^{H,in}$	0.0	$b_{4,2,2}^{H,out}$	0.0
$b_{1,5,1}^{H,in}$	0.0	$b_{5,4,1}^{H,in}$	0.0	$b_{4,3,1}^{H,out}$	0.0
$b_{1,5,2}^{H,in}$	0.0	$b_{5,5,1}^{H,in}$	0.0	$b_{4,4,1}^{H,out}$	0.0
$b_{2,1,1}^{H,in}$	0.0	$b_{5,5,2}^{H,in}$	0.0	$b_{4,4,2}^{H,out}$	0.0
$b_{2,1,2}^{H,in}$	0.0	$b_{1,1,1}^{H,out}$	1408.0	$b_{4,5,1}^{H,out}$	0.0
$b_{2,2,2}^{H,in}$	0.0	$b_{1,2,1}^{H,out}$	0.0	$b_{4,5,2}^{H,out}$	0.0
$b_{2,3,1}^{H,in}$	0.0	$b_{1,2,2}^{H,out}$	0.0	$b_{5,1,2}^{H,out}$	0.0
$b_{2,3,2}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	0.0	$b_{5,2,1}^{H,out}$	0.0
$b_{2,4,1}^{H,in}$	0.0	$b_{1,3,2}^{H,out}$	0.0	$b_{5,2,2}^{H,out}$	1656.5740269487962
$b_{2,5,1}^{H,in}$	1188.3798829882844	$b_{1,4,1}^{H,out}$	0.0	$b_{5,3,1}^{H,out}$	3262.333030757717
$b_{2,5,2}^{H,in}$	0.0	$b_{1,4,2}^{H,out}$	0.0	$b_{5,3,2}^{H,out}$	0.0
$b_{3,1,1}^{H,in}$	0.0	$b_{1,5,1}^{H,out}$	0.0	$b_{5,4,1}^{H,out}$	0.0
$b_{3,1,2}^{H,in}$	1531.3068929661752	$b_{1,5,2}^{H,out}$	0.0	$b_{5,5,1}^{H,out}$	0.0
$b_{3,2,1}^{H,in}$	1182.0343451272504	$b_{2,1,1}^{H,out}$	0.0	$b_{5,5,2}^{H,out}$	0.0
$b_{3,2,2}^{H,in}$	0.0	$b_{2,1,2}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	0.0
$b_{3,3,1}^{H,in}$	0.0	$b_{2,2,2}^{H,out}$	0.0	$b_{1,2,1}^{C,in}$	0.0
$b_{3,3,2}^{H,in}$	0.0	$b_{2,3,1}^{H,out}$	0.0	$b_{1,2,2}^{C,in}$	0.0
$b_{3,4,1}^{H,in}$	0.0	$b_{2,3,2}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	1857.24
$b_{3,4,2}^{H,in}$	279.5631070338243	$b_{2,4,1}^{H,out}$	0.0	$b_{1,3,2}^{C,in}$	0.0
$b_{3,5,2}^{H,in}$	0.0	$b_{2,5,1}^{H,out}$	1188.3798829882844	$b_{1,4,1}^{C,in}$	0.0
$b_{4,1,1}^{H,in}$	3415.86	$b_{2,5,2}^{H,out}$	0.0	$b_{1,4,2}^{C,in}$	0.0
$b_{4,1,2}^{H,in}$	0.0	$b_{3,1,1}^{H,out}$	0.0	$b_{1,5,1}^{C,in}$	0.0
$b_{4,2,1}^{H,in}$	0.0	$b_{3,1,2}^{H,out}$	1531.3068929661752	$b_{1,5,2}^{C,in}$	1296.87
$b_{4,2,2}^{H,in}$	0.0	$b_{3,2,1}^{H,out}$	1182.0343451272504	$b_{2,1,1}^{C,in}$	0.0
$b_{4,3,1}^{H,in}$	0.0	$b_{3,2,2}^{H,out}$	0.0	$b_{2,1,2}^{C,in}$	0.0
$b_{4,4,1}^{H,in}$	0.0	$b_{3,3,1}^{H,out}$	0.0	$b_{2,2,2}^{C,in}$	705.16

Variable	Value	Variable	Value	Variable	Value
$b_{2,3,1}^{C,in}$	0.0	$b_{1,2,2}^{C,out}$	0.0	$b_{5,1,2}^{C,out}$	0.0
$b_{2,3,2}^{C,in}$	0.0	$b_{1,3,1}^{C,out}$	1857.24	$b_{5,2,1}^{C,out}$	0.0
$b_{2,4,1}^{C,in}$	0.0	$b_{1,3,2}^{C,out}$	0.0	$b_{5,2,2}^{C,out}$	0.0
$b_{2,5,1}^{C,in}$	0.0	$b_{1,4,1}^{C,out}$	0.0	$b_{5,3,1}^{C,out}$	0.0
$b_{2,5,2}^{C,in}$	0.0	$b_{1,4,2}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{3,1,1}^{C,in}$	0.0	$b_{1,5,1}^{C,out}$	0.0	$b_{5,4,1}^{C,out}$	0.0
$b_{3,1,2}^{C,in}$	0.0	$b_{1,5,2}^{C,out}$	1296.87	$b_{5,5,1}^{C,out}$	0.0
$b_{3,2,1}^{C,in}$	0.0	$b_{2,1,1}^{C,out}$	0.0	$b_{5,5,2}^{C,out}$	0.0
$b_{3,2,2}^{C,in}$	0.0	$b_{2,1,2}^{C,out}$	0.0		
$b_{3,3,1}^{C,in}$	0.0	$b_{2,2,2}^{C,out}$	705.16		
$b_{3,3,2}^{C,in}$	0.0	$b_{2,3,1}^{C,out}$	0.0		
$b_{3,4,1}^{C,in}$	0.0	$b_{2,3,2}^{C,out}$	0.0		
$b_{3,4,2}^{C,in}$	0.0	$b_{2,4,1}^{C,out}$	0.0		
$b_{3,5,2}^{C,in}$	0.0	$b_{2,5,1}^{C,out}$	0.0		
$b_{4,1,1}^{C,in}$	0.0	$b_{2,5,2}^{C,out}$	0.0		
$b_{4,1,2}^{C,in}$	103.8239999999989	$b_{3,1,1}^{C,out}$	0.0		
$b_{4,2,1}^{C,in}$	0.0	$b_{3,1,2}^{C,out}$	0.0		
$b_{4,2,2}^{C,in}$	0.0	$b_{3,2,1}^{C,out}$	0.0		
$b_{4,3,1}^{C,in}$	0.0	$b_{3,2,2}^{C,out}$	0.0		
$b_{4,4,1}^{C,in}$	3056.91	$b_{3,3,1}^{C,out}$	0.0		
$b_{4,4,2}^{C,in}$	0.0	$b_{3,3,2}^{C,out}$	0.0		
$b_{4,5,1}^{C,in}$	0.0	$b_{3,4,1}^{C,out}$	0.0		
$b_{4,5,2}^{C,in}$	0.0	$b_{3,4,2}^{C,out}$	0.0		
$b_{5,1,2}^{C,in}$	0.0	$b_{3,5,2}^{C,out}$	0.0		
$b_{5,2,1}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	0.0		
$b_{5,2,2}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	103.8239999999989		
$b_{5,3,1}^{C,in}$	0.0	$b_{4,2,1}^{C,out}$	0.0		
$b_{5,3,2}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	0.0		
$b_{5,4,1}^{C,in}$	0.0	$b_{4,3,1}^{C,out}$	0.0		
$b_{5,5,1}^{C,in}$	0.0	$b_{4,4,1}^{C,out}$	3056.91		
$b_{5,5,2}^{C,in}$	0.0	$b_{4,4,2}^{C,out}$	0.0		
$b_{1,1,1}^{C,out}$	0.0	$b_{4,5,1}^{C,out}$	0.0		
$b_{1,2,1}^{C,out}$	0.0	$b_{4,5,2}^{C,out}$	0.0		